

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0932

Fudge Hollow 138 kV
June 20, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0932. The Interconnection Customer proposes a 280.5 MW battery energy storage facility interconnecting at the Fudge Hollow 138 kV substation in the Virginia area. The request was received on January 27, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 349 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 31 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$141,176,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lowmoor / Fudge Hollow 138 KV
Contingency: Aspen / Lowmoor 345 KV
Loading: 190.6% (Rating: 190 MVA)
DFAX: 85.8%
Pre-project loading: 64.3%
2. Facility: Imhoff Tap 1 / Ola 138 KV
Contingency: Lowmoor / Fudge Hollow Substation 138 KV
Loading: 163.8% (Rating: 181 MVA)
DFAX: 100.0%
Pre-project loading: 8.9%
3. Facility: Crockett Reit / Ola 138 KV Line
Contingency: Lowmoor / Fudge Hollow 138 KV
Loading: 154.6% (Rating: 201 MVA)
DFAX: 100.0%
Pre-project loading: 15.4%
4. Facility: Imhoff Tap 1 / Fudge Hollow Substation 138 KV L
Contingency: Lowmoor / Fuge Hollow 138 KV
Loading: 153.3% (Rating: 183 MVA)
DFAX: 100.0%
Pre-project loading: 0.0%
5. Facility: Superior Substation / Elmwest 138 KV
Contingency: Lowmoor / Fudge Hollow 138 KV
Loading: 151.2% (Rating: 223 MVA)
DFAX: 100.0%
Pre-project loading: 25.2%

6. Facility: Great Falls / Crockett Reit 138 KV
Contingency: Fudge Hollow / Lowmoor 138 KV Line
Loading: 150.7% (Rating: 210 MVA)
DFAX: 100.0%
Pre-project loading: 17.2%
7. Facility: Great Falls / Superior 138 KV Line
Contingency: Lowmoor / Fudge Hollow 138 KV
Loading: 150.4% (Rating: 220 MVA)
DFAX: 100.0%
Pre-project loading: 23.1%
8. Facility: Seymour / Havasu 138 KV Line
Contingency: Fudge Hollw / Lowmoor 138 KV
Loading: 139.3% (Rating: 248 MVA)
DFAX: 100.0%
Pre-project loading: 26.0%
9. Facility: W Cape / Seymour 138 KV
Contingency: Fudge Hollow / Lowmoor 138 KV
Loading: 139.1% (Rating: 247 MVA)
DFAX: 100.0%
Pre-project loading: 25.5%
10. Facility: West Cape / Elmwest 138 KV line
Contingency: Fudge Hollow / Lowmoor 138 KV L
Loading: 139.0% (Rating: 246 MVA)
DFAX: 100.0%
Pre-project loading: 24.8%
11. Facility: Michigan Power Lp / Osceola 138 KV
Contingency: Aspen / Lowmoor 345 KV line
Loading: 128.8% (Rating: 355 MVA)
DFAX: 14.0%
Pre-project loading: 117.8%
12. Facility: Osceola / Lone Tree 138 KV
Contingency: Lowmoor / Aspen 345 KV
Loading: 128.2% (Rating: 361 MVA)
DFAX: 14.0%
Pre-project loading: 117.3%
13. Facility: Lowmoor 345 kV/138 kV Auto
Contingency: Aspen / Lowmoor 345 KV
Loading: 125.3% (Rating: 534 MVA)
DFAX: 62.0%
Pre-project loading: 92.7%
14. Facility: Lowmoor / Talley Rd 138 KV
Contingency: Aspen / Lowmoor 345 KV line
Loading: 124.2% (Rating: 230 MVA)
DFAX: 9.8%
Pre-project loading: 112.2%
15. Facility: Clifton Forge / Talley Road 138 KV L
Contingency: Aspen / Lowmoor 345 KV
Loading: 123.8% (Rating: 226 MVA)
DFAX: 9.8%
Pre-project loading: 111.6%
16. Facility: Clifton Forge / Richmond Q 138 KV Line
Contingency: Lowmoor / Aspen Substation 345 KV

Loading: 122.8% (Rating: 219 MVA)

DFAX: 9.8%

Pre-project loading: 110.3%

17. Facility: Kattelville / Richmond Q 138 KV

Contingency: Aspen / Lowmoor 345 KV Line

Loading: 121.8% (Rating: 212 MVA)

DFAX: 9.8%

Pre-project loading: 108.9%

18. Facility: Trego / Beaver 500 KV line

Contingency: Adams / Onancock Diesel Generation Facility 230 KV line

Loading: 119.6% (Rating: 4,242 MVA)

DFAX: 7.8%

Pre-project loading: 119.1%

19. Facility: Craigsville / Yucaipa 138 KV

Contingency: Lowmoor / Aspen 345 KV

Loading: 118.3% (Rating: 205 MVA)

DFAX: 9.8%

Pre-project loading: 104.9%

20. Facility: Kattelville / Craigsville 138 KV L

Contingency: Lowmoor / Aspen 345 KV

Loading: 117.3% (Rating: 216 MVA)

DFAX: 9.8%

Pre-project loading: 104.5%

21. Facility: Wadsworth / Oak Crk Sta 138 KV

Contingency: Dayton / Chatom 345 KV

Loading: 115.9% (Rating: 328 MVA)

DFAX: 5.5%

Pre-project loading: 111.2%

22. Facility: Oak Crk Station / Sans Souci 138 KV Line

Contingency: Dayton / Chatom 345 KV

Loading: 112.5% (Rating: 372 MVA)

DFAX: 5.5%

Pre-project loading: 108.3%

23. Facility: Harrisonburg / Sans Souci 138 KV L

Contingency: Dayton / Chatom 345 KV

Loading: 112.5% (Rating: 372 MVA)

DFAX: 5.5%

Pre-project loading: 108.3%

24. Facility: Hinton Ap 345-138 kV TX

Contingency: Manatee / Hinton Ap 345 KV Line

Loading: 108.7% (Rating: 640 MVA)

DFAX: 17.4%

Pre-project loading: 101.1%

25. Facility: Galion Generating Sta / Trego 500 KV line

Contingency: Trego / Beaver 500 KV

Loading: 106.4% (Rating: 2,687 MVA)

DFAX: 5.3%

Pre-project loading: 105.9%

26. Facility: Redlnd / Carodean 138 KV

Contingency: Dayton / Crandall 345 KV

Loading: 104.6% (Rating: 368 MVA)

DFAX: 8.0%
Pre-project loading: 98.5%

27. Facility: Weyers Cave / Greenbank 138 KV L

Contingency: Crandall / Dayton 345 KV line
Loading: 104.0% (Rating: 398 MVA)
DFAX: 7.7%
Pre-project loading: 98.6%

28. Facility: Caodean / Crandall 138 KV

Contingency: Dyton / Crandall 345 KV
Loading: 103.2% (Rating: 400 MVA)
DFAX: 8.0%
Pre-project loading: 97.7%

29. Facility: Hinton Ap / Antelope Vly Station 138 KV

Contingency: Barr Lake / Kean Mtn And Grassy Creek 345 KV
Loading: 102.6% (Rating: 564 MVA)
DFAX: 9.9%
Pre-project loading: 97.7%

30. Facility: St Louis / Nichols 138 KV

Contingency: Page / Purdy 138 KV
Loading: 100.6% (Rating: 346 MVA)
DFAX: 5.2%
Pre-project loading: 96.3%

31. Facility: Crndall Substation 345-138 kV TX

Contingency: Dayton / Crandall 345 KV Line
Loading: 100.5% (Rating: 463 MVA)
DFAX: 6.9%
Pre-project loading: 96.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Fudge Hollow / Lowmoor 138 KV	Rebuild line with higher-capacity conductor	\$9,916,000	94% (\$9,271,000)
Imhoff Tap 1 / Ola 138 KV	Rebuild line with higher-capacity conductor	\$6,691,000	100% (\$6,691,000)
Crockett Reit / Ola 138 KV Line	Rebuild line with higher-capacity conductor	\$5,256,000	100% (\$5,256,000)
Imhoff Tap 1 / Fudge Hollow Substation 138 KV L	Rebuild line with higher-capacity conductor	\$7,674,000	100% (\$7,674,000)
Elmwest Substation / Superior 138 KV	Rebuild line with higher-capacity conductor	\$8,995,000	100% (\$8,995,000)
Great Falls / Crockett Reit 138 KV	Rebuild line with higher-capacity conductor	\$1,144,000	100% (\$1,144,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Great Falls Substation / Superior 138 KV	Rebuild line with higher-capacity conductor	\$8,602,000	100% (\$8,602,000)
Seymour / Havasu 138 KV Line	Rebuild line with higher-capacity conductor	\$7,575,000	100% (\$7,575,000)
W Cape / Seymour 138 KV	Rebuild line with higher-capacity conductor	\$9,039,000	100% (\$9,039,000)
Wes Cape / Elmwst 138 KV L	Rebuild line with higher-capacity conductor	\$9,791,000	100% (\$9,791,000)
Michigan Power Lp / Osceola 138 KV	Rebuild line with higher-capacity conductor	\$13,269,000	10% (\$1,392,000)
Osceola / Lone Tree 138 KV	Rebuild line with higher-capacity conductor	\$5,912,000	10% (\$620,000)
Lowmoor 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$23,210,000	26% (\$5,992,000)
Lowmoor / Talley Rd 138 KV	Rebuild line with higher-capacity conductor	\$8,939,000	48% (\$4,268,000)
Clifton Forge / Talley Road 138 KV L	Rebuild line with higher-capacity conductor	\$13,502,000	48% (\$6,446,000)
Clifton Forge / Richmond Q 138 KV	Rebuild line with higher-capacity conductor	\$2,549,000	48% (\$1,217,000)
Kattelville / Richmond Q 138 KV	Rebuild line with higher-capacity conductor	\$35,441,000	48% (\$16,921,000)
Trego / Beaver 500 KV line	Rebuild line with higher-capacity conductor	\$408,491,000	3% (\$13,877,000)
Craigsville / Yucaipa 138 KV	Rebuild line with higher-capacity conductor	\$9,205,000	48% (\$4,395,000)
Kattelville / Craigsville 138 KV L	Rebuild line with higher-capacity conductor	\$6,603,000	48% (\$3,152,000)
Wadsworth / Oak Crk Sta 138 KV	Rebuild line with higher-capacity conductor	\$6,942,000	16% (\$1,125,000)
Oak Crk Station / Sans Souci 138 KV Line	Rebuild line with higher-capacity conductor	\$1,803,000	16% (\$292,000)
Harrisonburg / Sans Souci 138 KV	Rebuild line with higher-capacity conductor	\$2,327,000	16% (\$377,000)
Hinton Ap 345-138 kV TX	Replace transformer with higher-rated unit	\$18,262,000	7% (\$1,204,000)
Galion Generating Sta / Trego 500 KV	Reconductor line	\$95,454,000	3% (\$2,534,000)
Redlnd / Carodean 138 KV	Reconductor line	\$510,000	13% (\$68,000)
Weyers Cave / Greenbank 138 KV L	Reconductor line	\$1,780,000	11% (\$199,000)
Crandall / Carodean 138 KV line	Reconductor line	\$764,000	13% (\$101,000)
Hinton Ap / Antelope Vly Station 138 KV	Reconductor line	\$6,981,000	4% (\$265,000)
St Louis / Nichols 138 KV	Reconductor line	\$1,720,000	12% (\$214,000)
Crandall 345/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$18,687,000	13% (\$2,479,000)
	Total allocated to project		\$141,176,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.